

BENSON, MN, USA

WMO: 727507

Lat: 45.332N

Lon: 95.651W

Elev: 1039

StdP: 14.15

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 54904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-11.2	-23.9	1.5	-14.4	-17.9	2.1	-9.6	32.0	11.2	28.6	7.4	5.9	350	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	90.3	72.5	87.8	71.8	84.0	69.8	77.1	85.5	74.8	82.9	72.9	81.0	12.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	134.3	82.3	72.3	124.1	80.4	70.2	115.7	78.6	41.2	85.6	39.0	82.7	37.1	81.1	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	23.3	19.9	-23.0	95.2	6.6	3.4	-27.8	97.7	-31.7	99.7	-35.4	101.6	-40.2	104.1
			-23.1	80.2	6.5	2.3	-27.8	81.8	-31.6	83.1	-35.2	84.4	-39.9	86.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.9	12.7	16.0	29.7	44.3	57.0	67.2	71.4	68.3	61.1	47.2	32.0	18.4
	DBStd	23.13	13.69	13.57	13.60	10.49	8.97	6.93	6.05	5.90	8.85	9.42	11.32	12.59
	HDD50	4711	1155	951	639	226	35	0	0	0	12	166	547	980
	HDD65	8258	1620	1371	1096	622	277	55	14	34	178	555	991	1445
	CDD50	2497	0	0	9	57	253	515	663	568	345	81	6	0
	CDD65	567	0	0	0	2	30	120	212	137	61	5	0	0
	CDH74	5951	0	0	2	56	438	1246	2178	1316	633	81	1	0
	CDH80	1886	0	0	0	16	144	392	753	369	192	20	0	0
Wind	WSAvg	8.6	9.2	9.4	9.6	10.1	9.7	8.0	6.6	6.4	7.6	9.0	9.1	8.7
Precipitation	PrecAvg	26.9	0.7	0.7	1.5	2.6	3.0	4.1	3.9	3.5	3.0	2.5	1.1	0.7
	PrecMax	36.3	1.8	1.5	3.8	7.1	5.9	7.2	7.6	6.6	6.0	6.7	3.5	2.0
	PrecMin	19.1	0.1	0.2	0.5	0.9	0.4	1.6	0.7	1.0	1.0	0.3	0.0	0.2
	PrecStd	4.3	0.5	0.4	0.9	1.5	1.3	1.5	1.8	1.6	1.5	1.8	0.9	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	53.6	70.1	81.1	90.7	91.5	94.6	91.1	90.4	82.1	67.6	49.5
		MCWB	37.9	46.2	54.8	58.3	65.7	72.5	74.9	75.4	69.7	61.8	53.1	39.9
	2%	DB	37.3	43.4	61.2	73.1	83.7	88.3	90.3	87.8	84.4	73.5	60.6	43.1
		MCWB	33.9	38.2	51.1	53.2	63.4	70.3	74.8	73.8	68.7	57.5	49.0	37.2
	5%	DB	35.2	37.4	54.2	67.9	79.1	84.1	87.8	83.8	81.4	68.0	53.7	37.3
		MCWB	32.8	34.4	47.4	51.8	60.4	67.8	73.8	71.4	67.0	57.2	46.0	34.1
	10%	DB	32.0	35.8	47.6	63.0	73.2	81.3	83.9	81.3	77.1	63.2	47.6	35.7
		MCWB	30.6	33.5	43.0	49.5	57.8	66.2	71.6	69.5	64.5	53.3	42.0	33.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.3	44.8	59.1	61.0	70.9	76.9	80.8	78.8	75.2	66.1	56.7	43.2
		MCDB	43.6	50.5	64.0	75.1	82.8	87.0	88.4	85.1	84.0	74.7	64.6	47.4
	2%	WB	34.6	38.9	52.7	57.1	66.8	73.8	77.9	76.1	72.5	61.8	50.3	37.7
		MCDB	37.5	43.0	59.1	68.4	77.9	83.0	86.8	83.6	81.0	70.7	58.1	42.0
	5%	WB	32.7	35.4	47.3	53.7	63.7	71.3	75.6	74.0	69.8	57.9	46.1	34.9
		MCDB	34.6	38.0	53.8	64.2	73.3	81.5	83.7	81.2	77.6	65.8	51.8	37.7
	10%	WB	30.1	32.8	42.4	50.7	60.6	68.8	73.4	71.7	67.0	54.4	42.1	32.6
		MCDB	31.5	35.2	47.8	60.6	70.6	77.7	81.5	78.9	74.8	61.8	47.8	35.0
Mean Daily Temperature Range	5% DB	MDBR	18.2	18.7	18.3	21.8	22.7	21.5	22.1	22.4	23.8	21.5	18.5	16.5
		MCDBR	20.2	19.7	27.1	32.1	30.3	25.5	24.3	24.7	28.0	30.0	27.8	21.0
		MCWBR	17.3	15.8	17.5	16.3	14.4	11.9	12.1	13.0	13.9	16.4	17.6	16.5
	5% WB	MCDBR	18.8	18.5	25.5	26.9	24.5	21.5	21.0	21.4	23.1	24.3	25.6	19.7
	MCWBR	16.4	14.8	17.6	16.0	13.8	12.1	12.3	13.2	13.9	15.2	17.3	15.8	
Clear-Sky Solar Irradiance	taub	0.264	0.278	0.308	0.363	0.388	0.394	0.394	0.403	0.365	0.316	0.281	0.258	
	taud	2.436	2.401	2.431	2.329	2.319	2.360	2.348	2.310	2.398	2.525	2.533	2.468	
	Ebn at Noon	274	290	295	284	279	277	276	268	271	273	266	265	
	Edh at Noon	22	28	31	38	39	38	38	38	32	24	20	19	
All-Sky Solar Radiation	RadAvg	556	899	1206	1500	1723	1918	2055	1709	1313	836	538	394	
	RadStd	68	102	111	129	130	125	73	141	97	117	52	47	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page